# SIMATS ENGINEERING

## ASSESSMENT TOOL 3: Short Answer Test

**COURSE NAME:** Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

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**COURSE OUTCOME COVERED**

**CO1:** *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 5

**Total Marks: 25**

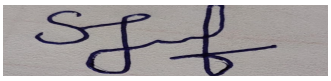
### SHORT ANSWER TEST

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**Code of Conduct**

I S Jenefar 192511345certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student:

**Assessment Tasks**

**Short Answer Test**

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Discuss the role of **data centers** in delivering cloud services..
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

**Short Answer Test Rubric**

| Criteria | Weightage | Level 4 - Exemplary | Level 3 - Proficient | Level 2 - Developing | Level 1 - Beginning |
|---|---|---|---|---|---|
| Conceptual Accuracy | 30% | Accurate definitions and precise explanation of all key terms | Mostly accurate with minor gaps | Partial understanding with noticeable errors | Incorrect or irrelevant explanation |
| Coverage of Concepts | 20% | Covers all expected sub-points completely | Covers most expected points | Covers only basic points | Very limited coverage |
| Use of Examples | 15% | Uses highly relevant cloud examples to strengthen the answer | Uses relevant examples with minor mismatch | Uses vague or partially relevant examples | No useful examples |
| Comparison / Differentiation | 20% | Clearly distinguishes concepts with strong logic | Reasonable differentiation with minor overlap | Comparison is weak or incomplete | Fails to differentiate concepts |
| Clarity of Expression | 15% | Clear, organized, and concise answer writing | Generally clear with few issues | Understandable but poorly organized | Difficult to follow |

SHORT ANSWER TEST
CO1 AT3

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Course : Cloud Computing
C.Code : CSA1520

01. Define cloud computing and explain any four characteristics that make it different from traditional on promise computing.

02. Trace the evolution of cloud computing from hardware to internet software evolution and show how these changes enabled modern cloud platforms.

03. Explain the role of server virtualization in cloud computing Differentiate b/w virtualization as an enabling technology and cloud computing as a servile model

04. Discuss the role of data centers in delivering cloud services

05. Differentiate Iaas, Paas, Saas & Xaas with one suitable real world example for each model.

## CLOUD COMPUTING :

Cloud computing is a delivery of computing service such as servers, storage, database, networking & software over internet on pay as you use basis, without needing to own or maintain physical infrastructure.

### FOUR CHARACTERISTICS OF CLOUD COMPUTING

01. On demand self Service - user can access computing resources whenever needed without directly contacting the service provider.

02. Broad Network Access - cloud services can be accessed through internet using devices such as laptops, Smart phones and tablets.

03. Resources Pooling - The provider shares computing resources such as servers and storage among multiple users based on their requirements

04. Rapid elasticity - Resources can quickly increased or decreased according to the demand.

## 02. EVOLUTION OF CLOUD COMPUTING :

cloud computing evolved gradually through improvements in hardware, networking, virtualization & internet software

only one organisation can use. Later powerful server & high speed processor were developed.

2. Distributed Computing : Computing task began to be divided among the multiple interconnected computer

3. Virtualisation : It allows one physical server to run multiple multiple virtual machines

4. Internet evolution : Growth of internet & broadband network made it possible access computing resources remotely.

5. Internet software evolution : Web technologies, Web services, APIs & service oriented architecture enabled application.

08. SERVER VIRTUALIZATION :

Role of server Virtualization

It is a technology that divide 1 physical server into multiple virtual machines. Each virtual machines can run it own OS & application independently.

1. Better utilization
2. Cost reduction
3. Scalability
4. Isolation
5. Easy Management

| VIRTUALIZATION | CLOUD COMPUTING |
|---|---|
| It is a enabling technology | Service Delivery model |
| Create virtual machine from physical resources | Provide computing resources through internet |
| Focus on resources sharing and isolation | Focus on on demand servie & accesibility |
| can exit without cloud computing | Commonly use virtualization as foundation. |
| Example: Running multiple VM on one server | Example: using cloud serva storage or software on demand |

04. DATA CENTERS

It is a facility that contain server, storage system, networking equipments, power and working system. If forms the physical foundation of cloud computing.

Roles of Data centers :

1. Resources Hosting : contain physical server & storage used to provide cloud services.

2. Data Storage : Store user file, database, backup secur

3. High availability : Multiple servers and backup syste help keep cloud service available

4. Scalability : Additional servers & storage added

5. Networking : High speed network connection allow use to access cloud servile from anywhere.

6. Security & Protection : Data enters use physical security, firwalls, access control & monitoring to protect resources & data.

05. CLOUD SERVICE :

Cloud services are divided based on what cloud provide manage and what the user manage.

Iaas → Infrastructure as a service

Provide virtualized computing infrastrastruture Such as server, storage & Networking.

Example:

Amazon EC2

Iaas => Get hardware resources

Paas ⇒ Platform as a service

Provide a readymade platform for development to build, test & deploy application.

Example:

Google App engine

Paas ⇒ Platform development

Saas ⇒ Software as a service

Provide a complete software application through internet.

Example:

Google doc

Saas ⇒ Ready made platform

Xaas ⇒ Anything as a Service

A broad from delivering different IT resources or service over internet such as database as a service, Storage as a service or security as a service.

Example:

Amazon RDS